

Observability

Can we reconstruct the state from the output?

Ogata, *Modern Control Engineering*, Ch. 9.

In this tutorial you will:

- test observability with the `obsv` rank condition and the PBH test,
- see the duality between observability and controllability, and
- watch a second sensor restore a lost mode.

Step through with **Ctrl+Enter**, or render a report with **publish**.

A system is **completely observable** if the initial state $x(0)$ can be determined from knowledge of the output $y(t)$ over a finite time interval, given the input $u(t)$. For LTI systems this holds iff the **observability matrix**

$$\mathbf{O} = [C; CA; CA^2; \dots; CA^{n-1}]$$

has full rank n .

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Example 1: An Observable System

```
A1 = [0 1; -2 -3];
C1 = [1 0];
Ob1 = obsv(A1,C1);
fprintf('Example 1: Observability matrix = \n')
disp(Ob1)
fprintf('rank = %d (n = %d) -> %s\n', rank(Ob1), size(A1,1), ...
    string(rank(Ob1)==size(A1,1)))
```

Example 1: Observability matrix =

```
1    0
0    1
```

rank = 2 (n = 2) -> true

Example 2: An Unobservable System

The sensor measures only a combination of states that is blind to one internal mode (a "hidden" state).

```
A2 = [-1 0; 0 -2];
C2 = [1 0];
Ob2 = obsv(A2,C2);
fprintf('\nExample 2: Observability matrix = \n')
disp(Ob2)
fprintf('rank = %d (n = %d) -> %s\n', rank(Ob2), size(A2,1), ...
    string(rank(Ob2)==size(A2,1)))
fprintf('State x2 has no effect on y: it is unobservable from this sensor.\n')
```

Example 2: Observability matrix =

```
1    0
-1   0
```

rank = 1 (n = 2) -> false

State x2 has no effect on y: it is unobservable from this sensor.

PBH Eigenvector Test for Observability

(A, C) is observable iff $\text{rank}[A - \lambda I; C] = n$ for every eigenvalue of A .

```
eigsA2 = eig(A2);
for i = 1:length(eigsA2)
    lam = eigsA2(i);
    PBH = [A2-lam*eye(2); C2];
    fprintf('lambda=%.2f: rank[A-lambda*I; C] = %d\n', lam, rank(PBH))
end
```

```
lambda=-2.00: rank[A-lambda*I; C] = 1
```

```
lambda=-1.00: rank[A-lambda*I; C] = 2
```

Duality Between Controllability and Observability

Kalman duality: (A, C) is observable iff (A^T, C^T) is controllable. This is exactly why observer gains are designed with ‘acker’/‘place’ applied to the transposed pair (see `StateObservers.m`).

```
Co_dual = ctrb(A1',C1');
Ob_direct = obsv(A1,C1);
fprintf('\nDuality check: ctrb(A^T,C^T)^T should equal obsv(A,C)\n')
fprintf('Match: %s\n', string(isequal(Co_dual', Ob_direct)))
```

```
Duality check: ctrb(A^T,C^T)^T should equal obsv(A,C)
```

```
Match: true
```

Example 3: Adding a Second Sensor Restores Observability

If a single output leaves a mode unobservable, adding measurements can recover full rank.

```
C2_aug = [1 0; 0 1];
Ob2_aug = obsv(A2,C2_aug);
fprintf('\nExample 2 with 2 outputs: rank(Ob) = %d (n=%d)\n', ...
        rank(Ob2_aug), size(A2,1))
```

```
Example 2 with 2 outputs: rank(Ob) = 2 (n=2)
```

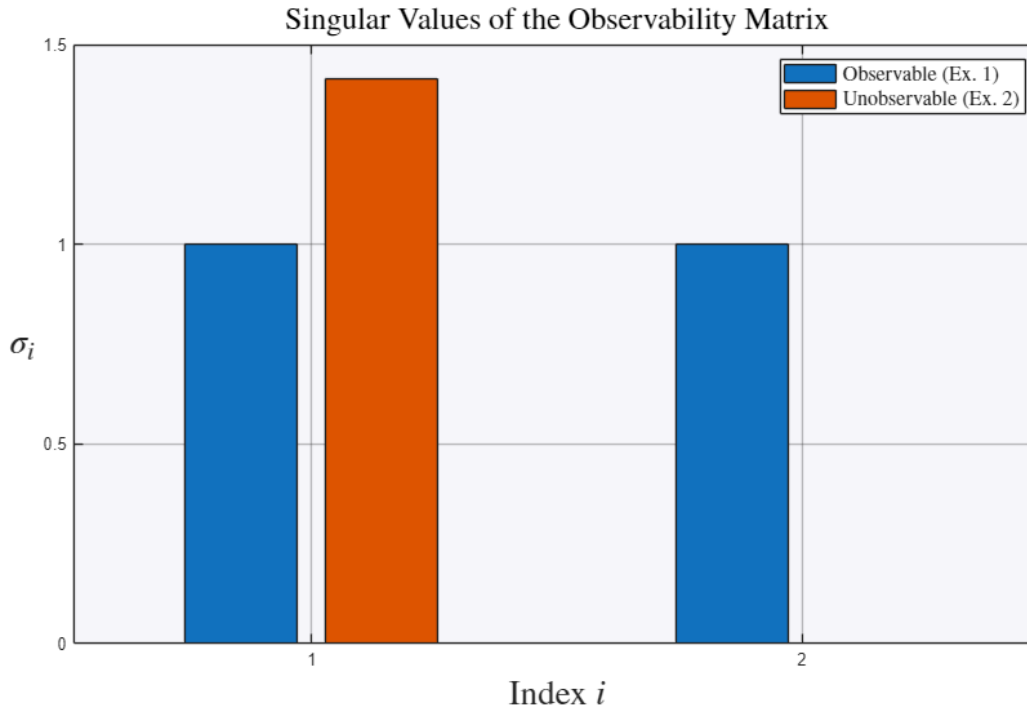
Observability and State Reconstruction

With (A, C) observable, an observer can be built to asymptotically reconstruct $x(t)$ from $y(t)$ and $u(t)$ alone, even though $x(t)$ itself is not directly measured – the foundation of `StateObservers.m`.

Visualizing the Rank Condition

Dually, the singular values of the observability matrix reveal the rank test: the unobservable system has a zero singular value, marking the hidden mode that no output combination exposes.

```
figure
bar([svd(Ob1), svd(Ob2)])
grid on
legend('Observable (Ex. 1)', 'Unobservable (Ex. 2)', 'Interpreter', 'latex', 'FontSize', 12)
title('Singular Values of the Observability Matrix', 'Interpreter', 'latex', 'FontSize', 18)
ylabel('\sigma_i', 'Interpreter', 'latex', 'FontSize', 20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('Index $i$', 'Interpreter', 'latex', 'FontSize', 20)
```



Try it yourself

- Change $C2$ to $[0 \ 1]$ and notice which mode becomes the hidden one.
- Check Kalman duality yourself: compare $\text{obsv}(A1, C1)$ with $\text{ctrb}(A1', C1')$.